

SAKTHI VEL C

+91 6374052055 | c.sakthivel1.3.2006@gmail.com

linkedin.com/in/sakthivel-c | github.com/sakthivel-136 | sakthivel-blog.io | leetcode.com/u/23ucs019

PROFESSIONAL SUMMARY

Computer Science student and Full-Stack Developer specializing in Machine Learning and IoT, focusing on architecting secure, scalable systems. Proven expertise in building real-time AI applications with Gemini 1.5 Pro, FastAPI, and Next.js, while optimizing database for performance and implementing robust encrypted data architectures for high-performance environments.

EDUCATION

Bachelor of Engineering in Computer Science

2023 – 2027

Kamaraj College of Engineering and Technology, Virudhunagar, India | GPA: 7.42/10

TECHNICAL SKILLS

Languages & Databases: Python, C, PostgreSQL, Supabase, SQL

Frontend & App: Next.js 14, Flutter, Tailwind CSS

Backend & APIs: FastAPI, Flask, RESTful APIs, JWT, Supabase Auth

AI & Machine Learning: Gemini 1.5 Pro, Random Forest, XGBoost, CNN, TensorFlow

IoT & Hardware: ESP Modules, Arduino IDE, MQTT, Sensor Integration

Security & Tools: Git, GitHub, RBAC, Jupyter Notebook

INTERNSHIP EXPERIENCE

Software Developer Intern

Dec 2025 – Present

PENTAGON GARMENTS, Virudhunagar, India

- Developed a secure verification system with RESTful APIs and JWT authentication for real-time security monitoring.

Machine Learning Intern

June 2025 – Nov 2025

PANITH INNOVATIONS, Remote

- Developed and deployed predictive Machine Learning models using XGBoost and Random Forest to enhance the system accuracy and project performance.

KEY PROJECTS

KURAL – AI Voice-Based Grievance Redressal System | Gemini 1.5 Pro, PSTN

JAN 2026 – Present

- Engineering a PSTN-based platform using Gemini 1.5 Pro to convert offline voice calls into structured digital reports.
- Implementing speech recognition and automated intent detection to classify and escalate grievances to authorities.

VANTAGE – AI Interview Intelligence Platform | Next.js 14, FastAPI, Mistral AI

JAN 2026

- Architected a full-stack AI interview platform featuring a 4-stage automated evaluation pipeline.
- Engineered real-time scoring using Mistral AI to evaluate response depth and contextual relevance.

Security Rounds Verifier | FastAPI, Next.js, PostgreSQL, JWT

JAN 2026

- Architected a full-stack verification system with Role-Based Access Control (RBAC) and real-time monitoring.
- Optimized database queries reducing response time by 40% for high-concurrency environments.

IoT Smart Air Quality & Climate Monitoring System | ESP32, Random Forest

Nov 2025

- Developed a predictive alert system using Random Forest to detect health risks and HVAC inefficiencies.
- Presented research findings at the ICASET-2026 International Conference in Chennai.

CONFERENCES & ACHIEVEMENTS

- Presenter** – 5th International Conference on Advances in Science, Engineering & Technology (ICASET-2026). Presented research on IoT-integrated Air Quality Systems.
- Patent Filed** – Intellectual Property application filed for "Mr.Strict" (AI Proctoring System).
- Grand Winner** – Student Pitch Competition at Mini Technovision 2K26.
- ISRO Certification** – Completed AIML training for Crop Acreage Mapping via official ISRO workshop.